

ORIGINAL ARTICLE



Item-validity analysis of the SED-S in a multicentre study of adults with intellectual disabilities

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Abstract

Background: Valid and reliable instruments for measuring emotional development are critical for a proper diagnostic assignment in individuals with intellectual disabilities. This exploratory study examined the psychometric properties of the items on the Scale of Emotional Development–Short (SED-S).

Method: The sample included 612 adults with intellectual disabilities ($M_{\text{age}} = 37.35$, $SD_{\text{age}} = 13.27$; 59.8% males). Item validity analysis comprising sensitivity and specificity rates and discriminatory power were determined.

Results: The relative mean frequency of ‘yes’ answers to all 200 items was 29.5%. The mean sensitivity rate was 67.5% and the mean specificity rate was 79.3%. Most items (85.0%) showed good discriminatory power with the adjacent stage(s), especially between SED-1, SED-2, SED-3 and SED-4. Particularly in SED-4 some items showed weaknesses in the differentiation between these stages.

Discussion: This study adds to previous validation studies by showing that most SED-S items have psychometrically sound properties.

KEYWORDS

challenging behaviour, emotional development, intellectual disabilities, neurodevelopmental disorders

1 | INTRODUCTION

Meta-analytic evidence shows that the prevalence of intellectual disabilities is approximately 1%–2% in the general population (Maulik et al., 2011). According to the International Classification of Diseases and Related Health Problems (ICD-11), disorders of intellectual development (in DSM-5 referred to as intellectual disabilities) comprises a group of etiologically diverse conditions characterised by significant impairments in intellectual functioning and adaptive functioning with onset during the developmental period (World Health Organization, 2018). In addition to intellectual and adaptive impairments, people with intellectual disabilities are more susceptible to experience co-occurring psychiatric disorders and somatic diseases (Mazza et al., 2020; Sappok, Diefenbacher, & Winterholler, 2019) and tend to have more frequent hospitalisations and a shorter life expectancy compared to the general population (O'Leary et al., 2018). Furthermore, individuals with intellectual disabilities are more likely to experience delays in their emotional development (ED), which in turn predict higher levels of challenging behaviours such as physical aggression and self-injury (Hermann et al., 2022; Sappok et al., 2014). Valid and reliable instruments for measuring ED are critical, not only to understand the behaviours of individuals with intellectual disabilities, but also to better tailor support and interventions to the needs of the individual. In the present pre-registered exploratory study, we examined the psychometric properties of the items on the Scale of Emotional Development–Short (SED-S; Sappok et al., 2016) in a sample of 612 adults with intellectual disabilities.

ED is a multifaceted concept encompassing different dimensions of socio-emotional functioning, including object permanency, relating to significant others, and differentiating and regulating emotions (de Bruijn et al., 2017, 2021; Došen, 2018; Sappok et al., 2021). In 2016, the SED-S was introduced as a short international scale to measure the level of ED in people with intellectual disabilities (Sappok et al., 2016). Other scales (SAED, SEO-Lukas-Version; SEO-R and SEO-R²) have been previously introduced and compared to measure the level of ED (Barrett & Kolb, 2015; Claes & Verduyn, 2012; Došen, 2005a, 2005b; Morisse & Došen, 2016; Sappok et al., 2023). The SED-S is based on Anton Došen's (Došen, 1990, 2014) conceptualisation of the developmental dynamic approach. In this approach, consecutive stages of ED are outlined which correspond to the normative trajectories in typically developing children and the concurrent maturation of associated brain circuits. In each stage, individuals display specific patterns of adaptive or maladaptive behaviours along with certain basic emotional needs which require fulfilment by the environment in order to achieve psychosocial homeostasis and to promote further development (Došen, 2005a, 2005b). Importantly, individuals with intellectual disabilities, like typically developing children, can progress through the different stages of ED, although progress may be delayed or incomplete (Došen, 1990, 2014; Greenspan, 1997; Sappok et al., 2022).

The SED-S includes 200 binary items (yes/no), with 5 items per domain (eight domains in total) across five developmental stages (Sappok et al., 2016). Participants receive eight domain scores and

one overall score for ED. More specifically, domain scores are determined by selecting the SED-S stage wherein a particular domain received the highest number of 'yes' answers. The overall score for ED is determined by ranking the domain stages from lowest to highest and then selecting the fourth lowest SED-S stage. SED-S has been normed in a sample of 160 typically developing children aged 0–12 (Sappok, Böhm, et al., 2019; Sappok, Diefenbacher, & Winterholler, 2019). For most items, the highest response probabilities were found in the target group as compared to the adjacent age groups. Furthermore, agreement between the classifications of the eight SED-S domains and children's respective chronological age was high as well as the agreement between SED-S overall scores and children's chronological age group (Sappok, Böhm, et al., 2019; Sappok, Diefenbacher, & Winterholler, 2019). Furthermore, Sappok et al. (2020), developed five tri-lingual and cross-cultural index cases (i.e., one for each level of ED) to standardise the application of the scale. In another study, the reliability and validity of the SED-S was tested in a sample of 118 children between 3 and 18 years old with intellectual disabilities ranging from profound to mild (Sterkenburg et al., 2021). Results indicated excellent internal consistency ($\alpha = .94$) of the inter-domains of the SED-S. In addition, the SED-S showed good convergent validity as overall and domain scores were negatively associated with severity of intellectual disabilities. In addition, divergent validity was indicated by non-significant associations between SED-S overall scores and chronological age as well as by partly independent associations between SED-S overall scores and adaptive functioning (Sterkenburg et al., 2021). Finally, the SED-S has been validated in a sample of 724 adults with intellectual disabilities from three European countries (Flachsmeyer et al., 2023). The scale demonstrated excellent internal consistency for the complete sample ($\alpha = .93$) and different subgroups (range $\alpha = .87$ – $.94$) (e.g., subgroups by level of intellectual disability or gender). Moreover, in line with Sterkenburg et al. (2021), the eight SED-S domains represented a unidimensional concept (Flachsmeyer et al., 2023). Altogether, these findings support the validity of the SED-S on domain and scale level.

Although studies validating the SED-S on domain and scale level are unquestionably important, they do not offer any information on the psychometric properties of the items on the scale. Research examining the item level properties may offer new perspectives and direct implications for future use and analysis of the scale (Carpenter et al., 2016). For instance, item level analyses can provide insight into patterns of endorsement across participants in different SED-S stages. Potentially, it can be shown that specific items are endorsed by all individuals in a particular stage but not by individuals in another stage, which would indicate that these items have good psychometric properties. It may also become apparent that item contents only rarely match by individuals in a particular stage or that items are applicable by all individuals, regardless of their assigned stage. These latter items may require removal or revision to further improve the application of the scale. Additionally, as the domain scores and overall score for ED are intended to inform treatment and intervention decisions, a rigorous evaluation of the item level properties is critical to provide high quality clinical care.

In sum, the aim of the present pre-registered study was to explore the psychometric properties of the items on the SED-S in a sample of adults with intellectual disabilities. More specifically, we examined the following research questions: (1) What is the frequency of 'yes' answers to each item on the scale, (2) What are the sensitivity and specificity rates of each item on the scale, and (3) What is the discriminatory power of each item on the scale? Given the exploratory nature of the study, no hypotheses were formulated.

2 | METHODS

2.1 | Participants

The sample consisted of 612 adults with intellectual disabilities (59.8% males) aged between 18 and 72 years old ($M = 37.35$, $SD = 13.27$). Level of intellectual disability, assessed by the Disability Assessment Schedule (DAS), ranged from mild (27%), moderate (35.6%), and severe (28.6%) to profound (8.8%). Most of the sample was diagnosed with an additional disorder (81.7%), of which autism spectrum disorder (ASD) is the most common (28.6%). For a more detailed description of the participants see Table 1. Individuals diagnosed with dementia, or an amnesic disorder were excluded from the study.

2.2 | Measurements

2.2.1 | Scale of Emotional Development—Short (SED-S)

The SED-S was administered to measure the level of ED (Sappok et al., 2016). The SED-S is a semi-structured interview consisting of 200 binary ('yes' or 'no') items assessing observable behaviours exhibited by the participant in the 2 weeks preceding the interview. The items are organised into eight domains: Relating to His/Her Own Body ('Body'), Relating to Significant Others ('Others'), Dealing with-Object Permanence ('Object'), Differentiating Emotions ('Emotions'), Relating to Peers ('Peers'), Engaging with the Material World ('Material'), Communicating with Others ('Communication'), and Regulating Affect ('Affect'). Each domain includes 25 items which correspond to five developmental stages with reference ages ranging from 0 to 12 years of age, thus each stage includes 5 items per domain and 40 items in total. The five stages are: *Adaptation* (0–6 months), *Socialisation* (7–18 months), *Individuation* (19–36 months), *Identification* (4–7 years) and *Reality Awareness* (8–12 years). The score for each domain corresponds to the stage with the highest number of 'yes' items. If two stages had the same number of 'yes' answers, the lower level was selected. Subsequently, the domain-specific results were ranked from lowest to highest and the fourth lowest domain stage indicated the overall level of ED (Sappok et al., 2016). The semi-structured interviews lasted between 30 and 60 min and were conducted by trained and independent professionals (mostly

TABLE 1 Description of the sample, total sample size $N = 612$.

Characteristics	n (%)
Level of intellectual disability	
Mild (ICD-10: F70)	165 (27.0)
Moderate (ICD-10: F71)	218 (35.6)
Severe (ICD-10: F72)	175 (28.6)
Profound (ICD-10: F73)	54 (8.8)
Additional disorders	
Disorders due to the use of psychoactive substances (F1x.x)	9 (1.5)
Schizophrenia, schizotypal and delusional disorders (F2x.x)	63 (10.3)
Affective disorders (F3x.x)	75 (12.3)
Neurotic, stress-related and somatoform disorders (F4x.x)	59 (9.6)
Disorders of adult personality and behaviour (F6x.x)	16 (2.6)
Autism spectrum disorders (F84.x)	175 (28.6)
Significant impairment of behaviour requiring attention or treatment (F7x.1)	269 (44.0)
No disorder	118 (19.3)
Level of emotional development	
Adaptation (SED-S 1)	90 (14.7)
Socialisation (SED-S 2)	147 (24.0)
Individuation (SED-S 3)	223 (36.4)
Identification (SED-S 4)	134 (21.9)
Reality awareness (SED-S 5)	18 (2.9)

Note: Multiple disorders may be present in one person. Disorders were determined according to the International Classification of Diseases and Related Health Problems, 10th edition (ICD-10). Fxx.x refer to codes and categories from the ICD-10.

Abbreviation: SED-S, Scale of Emotional Development—Short.

psychologists) with at least two informants (i.e., family members or close caregivers).

2.2.2 | Disability Assessment Schedule (DAS)

The Disability Assessment Schedule (DAS) was administered to measure the severity of intellectual disability (Holmes et al., 1982). The DAS is an informant-based structured interview consisting of 16 items assessing adaptive behaviours in four areas: (1) continence, (2) self-help, (3) communication and (4) cultural techniques. The scores of each item were summed to a total score (range 15–71) to determine the severity of intellectual disability (i.e., mild, moderate, severe, and profound).

2.3 | Setting and design

This methodological instrument validation study investigated the psychometrical properties of the items included in the SED-S. Data were

collected between March 2016 and November 2020 in two hospitals, four home care facility centres, residential locations and day care centres in Belgium ($n = 49$), Germany ($n = 518$), and the Netherlands ($n = 52$). Organisations that participated in the present study were: Tordale in Torhout (Belgium), the Evangelisches Krankenhaus Königin Elisabeth Herzberge in Berlin (Germany), St. Lukas-Klink in Liebenau (Germany), Cordaan in Amsterdam (the Netherlands), ORO in Helmond (the Netherlands), Bartiméus in Doorn (the Netherlands), and De Twentse Zorgcentra in Losser (the Netherlands). The study was pre-registered on the Open Science Framework (OSF; see: <https://osf.io/4n3kz>). Adaptations to the analysis plan can be found in a supplementary addendum.

2.4 | Ethics

All participating organisations received ethical approval prior to data collection. For Belgium, the University of Ghent ethics commission of the Faculty of Psychology and Pedagogic Science provided permission. For Germany, the ethics commission of the Charité university hospital in Berlin (Ethics vote: EA2/193/16) and the ethics committee of Stiftung Liebenau approved the study. For the Netherlands, each organisation received ethical approval from its own ethics commission (Bartiméus 30.11.2017; De Twentse Zorgcentra 10.09.2018; Cordaan 28.08.2017; ORO 24.05.2017). The present study was conducted according to the principles of the Declaration of Helsinki (World Medical Association, 2013). Participants and/or their legal guardians signed for informed consent. In some cases, informed consent was given by signing the contract with the hospital according to the Landeskrankehausgesetz for retrospective data analysis in routine patient care.

2.5 | Statistical analyses

Statistical analyses were performed in SPSS version 28, Excel version 16 and R version 4.2.2. First, relative frequency rates were calculated to determine how often each item received a 'yes' answers. Next, relative frequency rates were calculated to determine the sensitivity and specificity rates for each item on the scale. More specifically, we examined how often each item received a 'yes' answer from participants in the awarded domain stage (i.e., sensitivity rate) and how often each item received a 'no' answer from participants outside the respective domain stage (i.e., specificity rate). Finally, Chi-square tests were used to assess the discriminatory power of the items, that is, we examined which items received more 'yes' answers from participants in the respective domain stage than from participants in the adjacent domain stage(s). Of note, there was only a one-side upward or a one-side downward comparison if the domain stage was SED-S 1 or SED-S 5 and there were two one-sided comparisons (i.e., one upward and one downward comparison) if the domain stages were SED-S 2, SED-S 3 or SED-S 4. Participants with incomplete SED-S data were excluded from the analyses ($n = 13$). Given the exploratory nature of

the study, we approach the interpretation of p -values in an exploratory manner.

3 | RESULTS

3.1 | Preliminary analyses

A participants' stage of ED ranged from SED-S stage 1–5 with Stage 3 'Individuation' being most frequently recorded (36.3%). Stage 1 'Adaptation' and stage 5 'Reality awareness' showed the lowest prevalence rates, with 14.8% and 2.9%, respectively. Stage 2 'Socialisation' and Stage 4 'Identification' showed prevalence rates of 24% and 21.7%, respectively. In Figure 1, we present the distribution of participants across the eight domains and SED-S overall level. A non-parametric Dunnett's test, a multiple comparison procedure, was used to compare the eight domain scores with the SED-S overall score. Results indicated differences between the domain scores and the SED-S overall score ($p < .001$), except for the domain 'Others' ($p = .604$). The domains 'Body', 'Object', 'Material' and 'Communication' showed higher domain scores than the SED-S overall score, while the domains 'Emotion', 'Peers' and 'Affect' showed lower domain scores than the SED-S overall score (see Supplementary Table 1). In the next steps, we therefore examined the sensitivity and specificity rates and the discriminatory power of the SED-S items in relation to the domain scores rather than the overall SED-S score.

3.2 | Relative frequencies

Relative frequency rates of 'yes' answers to the 40 items in each SED-S stage were as follows: SED-S 1 ($M = 20.79\%$, $SD = 7.44$, range = 6–36), SED-S 2 ($M = 31.88\%$, $SD = 15.23$, range = 11–64), SED-S 3 ($M = 41.04\%$, $SD = 16.27$, range = 12–72), SED-S 4 ($M = 32.97\%$, $SD = 14.66$, range = 9–71), and SED-S 5 ($M = 20.55\%$, $SD = 12.12$, range = 4–51). The relative frequency rate of 'yes' answers to all 200 items on the scale was on average 29.5% ($SD = 15.51$, range = 4–72). For a more detailed description of the relative frequency rates of 'yes' answers see Tables 2–6.

3.3 | Sensitivity and specificity rates

Figure 2 presents the relative frequencies of 'yes' answers to the items in SED-S 1. Items received relatively more 'yes' answers from participants in the respective domain stage than from participants in a different domain stage. For example, the first item in the SED-S 1 domain 'Body' received a 'yes' answer from 76% of the participants in domain stage SED-S 1, whereas this item was answered with 'yes' by 54.8% of the participants in domain stage SED-S 2 and by 17.5% of the participants in domain stage SED-S 3. Similar patterns of responses were observed for the other SED-S stages.

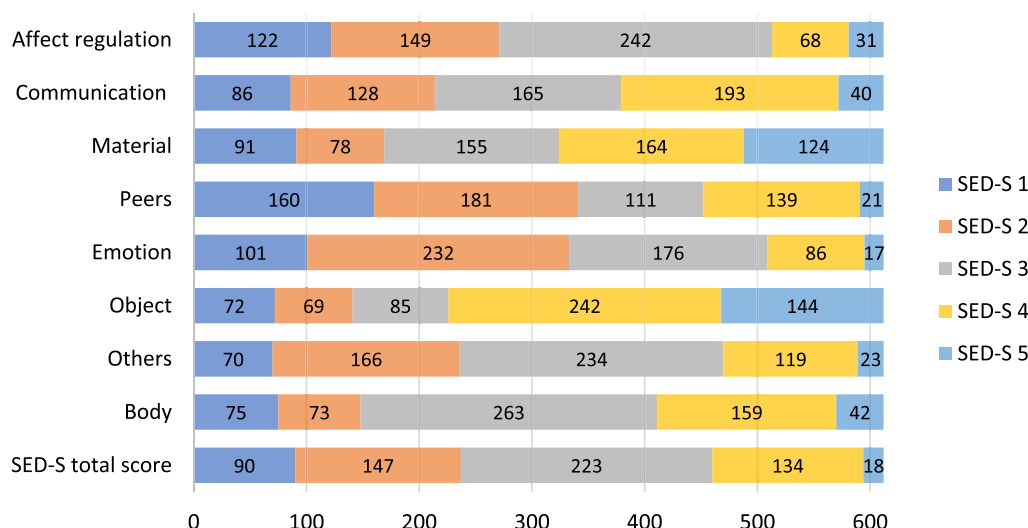


FIGURE 1 Distribution of participants across the eight domains and SED-S overall level. Example for interpretation: 122 participants were assigned to SED-S 1 in the domain 'Affect regulation', whereas 90 participants were assigned to an overall level of SED-S 1.

The sensitivity rates of the items in each stage were as follows: SED-S 1 ($M = 65.13\%$, $SD = 15.91$, range = 24–96), SED-S 2 ($M = 64.65\%$, $SD = 16.48$, range = 30–92), SED-S 3 ($M = 66.06\%$, $SD = 16.91$, range = 26–91), SED-S 4 ($M = 68.65\%$, $SD = 18.16$, range = 29–97) and SED-S 5 ($M = 71.92\%$, $SD = 18.04$, range = 19–93). The sensitivity rate for all 200 items on the scale was on average 67.3% ($SD = 17.16$, range 19–97). Furthermore, the number of items with a sensitivity rate above 70% increased per stage, SED-S 1; $n = 17/40$ (42.5%), SED-S 2; $n = 18/40$ (45.0%), SED-S 3; $n = 19/40$ (47.5%), SED-S 4; $n = 23/40$ (57.5%), SED-S 5; $n = 25/40$ (62.5%).

The specificity rates of the items in each stage were as follows: SED-S 1 ($M = 87.41\%$, $SD = 6.57$, range = 75–99), SED-S 2 ($M = 77.06\%$, $SD = 13.21$, range 50–95), SED-S 3 ($M = 69.53\%$, $SD = 14.93$, range = 37–91), SED-S 4 ($M = 77.82\%$, $SD = 12.79$, range 43–93), and SED-S 5 ($M = 84.63\%$, $SD = 9.17$, range = 62–97). The specificity rate for all 200 items on the scale was on average 79.3% ($SD = 13.21$, range 37–99). The number of items with a specificity rate above 70% was highest in SED-S 1 ($n = 40/40$; 100%), followed by SED-S 5 ($n = 37/40$; 92.5%), SED-S 4 ($n = 34/40$; 85.0%), SED-S 2 ($n = 29/40$, 72.5%), and SED-S 3 ($n = 23/40$; 57.5%). For a more detailed description of the sensitivity and specificity rates of the items in each SED-S stage, see Tables 2–6.

3.4 | Discriminatory power

Chi-square tests were used to assess the discriminatory power of the items on the scale. Overall, 90% of items discriminated significantly between adjacent stages.

For SED-S 1, most items (97.5%) received significantly more 'yes' answers from participants in the respective domain stage than from participants in the adjacent domain stage SED-S 2. For SED-S 2, all 40 items (100%) received significantly more 'yes' answers from

participants in the respective domain stage than from participants in either (SED-S-1, SED-3) of the adjacent domain stages. For SED-S 3, 39 out of 40 items (97.5%) and 34 out of 40 items (85%) received significantly more 'yes' answers from participants in the respective domain stage than from participants in the adjacent domain stages SED-S 2 and SED-S 4, respectively. For SED-S 4, 38 out of 40 items (95%) and 24 out of 40 items (60%) received significantly more 'yes' answers from participants in the respective domain stage than from participants in the adjacent domain stages SED-S 3 and SED-S 5, respectively. With respect to SED-S 4, results showed that the domain 'Peers' included only 1 item (out of 5) and the domain 'Communication' included only 2 items (out of 5) with sufficient power to discriminate between the response patterns of participants in the respective domain stage and participants in domain stage SED-S 5. Finally, for SED-S 5, most items (87.5%) received significantly more 'yes' answers from participants in the respective domain stage than from participants in the adjacent domain stage SED-S 4. Tables 2–6 show the individual results.

4 | DISCUSSION

The present pre-registered exploratory study examined the psychometric properties of the SED-S in a sample of adults with intellectual disabilities. More specifically, we explored the frequency of 'yes' answers, the sensitivity and specificity rates, and the discriminatory power of each item on the scale. Results indicated that most items on the scale have adequate sensitivity and specificity rates and satisfactory discriminatory power, indicating that the SED-S is a psychometric sound instrument to assess emotional development in persons with intellectual disabilities.

This study showed that the relative frequency rate of 'yes' answers to all 200 items on the scale was on average 29.45% and

TABLE 2 Overview of item validity assessment for SED-S 1 items.

SED-S 1 items	Domain and item number	Frequency rate of overall 'yes' answers	Relative frequency rate of 'yes' answers from participants in the respective domain stage					Specificity rate	Chi ² -test SED-S 1 versus 2 p-value
			SED-S (sensitivity)	SED-S 2	SED-S 3	SED-S 4	SED-S 5		
Emotional states are largely determined by basic physical sensations and needs (hunger, thirst, pain, fatigue, and cold)	Body 1_1	25.8%	76.0%	54.8%	17.5%	8.8%	2.4%	81.2%	.007
Only feels safe and secure in a familiar environment, that is, when surrounded by familiar faces and stimuli (e.g., touch, smells, sounds, etc.)	Body 1_2	19.6%	73.3%	32.9%	9.9%	6.9%	9.5%	87.9%	<.001
Passively enjoys sensory stimulation (e.g., being cared or touched)	Body 1_3	25.0%	74.7%	45.2%	18.3%	9.4%	2.4%	81.9%	<.001
Explores his/her body randomly by touching, grasping, (thumb) sucking, and so forth	Body 1_4	9.3%	45.3%	5.5%	4.9%	2.5%	4.8%	95.7%	<.001
Engages with his/her body by means of repetitive movements (flapping arms, rocking back and forth, etc.) and vocalisations	Body 1_5	23.9%	73.3%	42.5%	17.9%	8.2%	0.0%	83.1%	<.001
Social interaction mainly arises when basic needs (e.g., food, hygiene, and touch) are being met	Others 1_1	21.6%	82.9%	25.3%	13.2%	0.8%	0.0%	86.3%	<.001
Is soothed by physical contact with significant others (e.g., hugging, rocking)	Others 1_2	20.1%	57.1%	27.7%	12.8%	4.2%	8.7%	84.7%	<.001
Primarily interacts with his/her environment via the proximal senses (e.g., touch, smell and taste)	Others 1_3	11.8%	48.6%	14.5%	5.1%	0.8%	4.3%	93.0%	<.001
Enjoys feeling the spatial boundaries of his/her body through intensive physical contact with significant others (e.g., being held, caresses, massages, rocking, and tickling)	Others 1_4	21.4%	52.9%	27.1%	17.9%	4.2%	8.7%	82.7%	<.001
Contact is more difficult when there is a lot of commotion or noise (sensory overload)	Others 1_5	29.1%	67.1%	41.6%	22.6%	5.0%	13.0%	75.8%	<.001
Emotional states are primarily determined by the immediate situation	Object 1_1	24.0%	86.1%	43.5%	24.7%	12.4%	2.8%	84.3%	<.001
The environment is perceived solely through the senses	Object 1_2	13.9%	68.1%	18.8%	9.4%	5.0%	2.1%	93.3%	<.001
Persons or object that cannot be perceived using any sense no longer exist	Object 1_3	6.5%	51.4%	1.4%	0.0%	0.4%	0.7%	99.4%	<.001
Does not look for hidden/lost objects ('out of sight, out of mind')	Object 1_4	9.2%	63.9%	7.2%	2.4%	0.8%	0.7%	98.1%	<.001
Lives in the 'here and now'	Object 1_5	21.6%	95.8%	30.4%	15.3%	10.3%	2.8%	88.3%	<.001
Emotional states are expressed with the entire body	Emotion 1_1	34.2%	81.2%	40.9%	15.9%	3.5%	5.9%	75.1%	<.001
The expression of emotions changes rapidly in response to stimuli	Emotion 1_2	25.8%	62.4%	31.0%	11.4%	3.5%	0.0%	81.4%	<.001
Emotions are expressed intensely with no variation in degree	Emotion 1_3	20.1%	67.3%	17.2%	7.4%	2.3%	0.0%	89.2%	<.001
Reacts to sensory overload by retreating or shutting out stimuli (e.g., covering his/her ears)	Emotion 1_4	23.0%	69.3%	21.1%	9.1%	7.0%	0.0%	86.1%	<.001
Responds to aversive stimuli with a mix of tension, anxiety and anger	Emotion 1_5	29.1%	68.3%	33.6%	14.2%	7.0%	0.0%	78.7%	<.001
Shows no interest in peers	Peers 1_1	17.8%	60.0%	5.5%	2.7%	0.0%	0.0%	97.1%	<.001
Sees others as objects rather than beings who act and react	Peers 1_2	22.4%	58.8%	16.6%	10.8%	0.0%	4.8%	90.5%	<.001
Interacts with peers according to fixed patterns	Peers 1_3	18.0%	40.0%	18.2%	10.8%	0.7%	0.0%	89.8%	<.001

TABLE 2 (Continued)

SED-S 1 items	Domain and item number	Frequency rate of overall 'yes' answers	Relative frequency rate of 'yes' answers from participants in the respective domain stage						Chi ² -test SED-S 1 versus 2
			SED-S (sensitivity)	SED-S 2	SED-S 3	SED-S 4	SED-S 5	Specificity rate	
Actions elicit responses from peers	Peers 1_4	30.4%	51.9%	34.8%	27.0%	5.0%	14.3%	77.2%	.001
Forced contact with peers results in noticeable tension	Peers 1_5	22.1%	50.0%	18.8%	16.2%	2.2%	0.0%	87.8%	<.001
Is mainly preoccupied with his/her own body									
Only engages with materials and objects within immediate reach	Material 1_1	12.9%	64.8%	14.1%	5.8%	0.0%	0.0%	96.2%	<.001
Primarily engages in repetitive activities that stimulate the senses (e.g., rocking, making sounds, and touching)	Material 1_2	10.9%	49.5%	12.8%	7.7%	0.0%	0.0%	95.8%	<.001
	Material 1_3	23.4%	86.8%	30.8%	23.9%	1.2%	0.8%	87.7%	<.001
Enjoys handling and exploring amorphous materials such as water, sand or soap bubbles	Material 1_4	14.9%	51.6%	24.4%	11.6%	1.8%	3.2%	91.6%	<.001
Explores things by feeling, licking and tasting them	Material 1_5	15.5%	49.5%	21.8%	16.8%	3.0%	1.6%	90.4%	<.001
Undirected expression of inner states using the entire body	Communication 1_1	19.1%	79.1%	26.6%	5.5%	3.1%	0.0%	90.7%	<.001
Verbal communication and comprehension are lacking									
	Communication 1_2	6.2%	31.4%	7.0%	1.2%	0.0%	0.0%	97.9%	<.001
Conveys emotional states by crying, shrieking, clinging to caregivers, pacing back and forth	Communication 1_3	26.0%	82.6%	42.2%	12.1%	7.3%	0.0%	83.3%	<.001
Imitates facial expressions, sounds and words	Communication 1_4	10.0%	24.4%	18.8%	6.7%	2.6%	0.0%	92.4%	.318
Keeps repeating the same words and sounds	Communication 1_5	21.9%	72.1%	35.2%	14.5%	1.6%	0.0%	86.3%	<.001
Tension is relieved by (active or passive) movement and physical touch	Affect 1_1	26.5%	72.1%	24.8%	14.0%	1.5%	6.5%	84.9%	<.001
Responds to physical sensations and external stimuli with tension	Affect 1_2	33.0%	82.8%	36.2%	18.2%	1.5%	6.5%	79.4%	<.001
Uses repetitive movements (e.g., rocking, flapping arms, hitting, etc.) as a means of self-regulation	Affect 1_3	25.7%	74.6%	20.8%	14.5%	0.0%	0.0%	86.5%	<.001
Has no control over his/her impulses									
Displays auto-aggressive behaviour	Affect 1_4	24.2%	75.4%	21.5%	9.1%	2.9%	0.0%	88.6%	<.001
	Affect 1_5	35.8%	82.8%	32.9%	24.8%	10.3%	6.5%	75.9%	<.001
Mean		20.8%	65.1%	25.7%	12.6%	3.7%	2.8%	87.4%	

Note: Sensitivity and specificity rates < 70% are marked in bold. In each SED-S 1 domain, the item with the highest specificity rate and a *p*-value below .05 is marked in bold.

TABLE 3 Overview of validity assessment for SED-S 2 items.

SED-S 2 items	Domain and item number	Frequency rate of overall 'yes' answers	Relative frequency rate of 'yes' answers from participants in the respective domain stage					Specificity rate	Chi ² -test SED-S 2 versus 1	
			SED-S 1	SED-S 2 (sensitivity)	SED-S 3	SED-S 4	SED-S 5		p-value	
Uses mouth as well as hands to explore his/her environment	Body 2_1	14.9%	29.3%	54.8%	8.4%	3.8%	2.4%	90.5%	.002	<.001
Uses his/her body as an instrument to explore the immediate environment (e.g., switching the light on and off repeatedly, etc.)	Body 2_2	30.4%	25.3%	78.1%	30.0%	16.4%	11.9%	76.1%	<.001	<.001
Smears faeces and body fluids (saliva, blood, and sperm)	Body 2_3	13.6%	28.0%	57.5%	6.8%	1.3%	0.0%	92.4%	<.001	<.001
Uses his/her body to manipulate objects in the immediate environment	Body 2_4	19.0%	24.0%	74.0%	12.9%	6.3%	0.0%	88.5%	<.001	<.001
Body-focused attention (e.g., during bathing, brushing hair) leads to pleasurable interactions	Body 2_5	31.2%	32.0%	71.2%	30.4%	18.9%	11.9%	74.2%	<.001	<.001
Sense of wellbeing is dependent on the presence of significant others	Others 2_1	36.1%	27.1%	78.3%	22.6%	12.6%	17.4%	79.6%	<.001	<.001
Shows a preference for contact with significant others	Others 2_2	59.3%	57.1%	92.2%	55.1%	31.1%	17.4%	52.9%	<.001	<.001
Stays close to significant others in unfamiliar surroundings	Others 2_3	51.0%	34.3%	83.1%	47.0%	28.6%	26.1%	61.0%	<.001	<.001
Protests when contact with significant others is lost or ended	Others 2_4	20.4%	11.4%	46.4%	14.5%	4.2%	4.3%	89.2%	<.001	<.001
Actively seeks physical proximity to significant others and follows them like a shadow	Others 2_5	32.4%	10.0%	67.5%	25.2%	14.3%	13.0%	80.7%	<.001	<.001
Is tense and upset when separated from significant others	Object 2_1	15.2%	13.9%	58.0%	15.3%	7.4%	8.3%	90.2%	<.001	<.001
Significant others provide a sense of security in unfamiliar situations	Object 2_2	45.1%	55.6%	85.5%	55.3%	41.3%	20.8%	60.0%	<.001	<.001
Briefly looks for things that have disappeared from sight	Object 2_3	22.7%	27.8%	62.3%	20.0%	19.4%	8.3%	82.3%	<.001	<.001
Enjoys playing peek-a-boo and hide-and-seek	Object 2_4	10.5%	8.3%	49.3%	9.4%	4.5%	3.5%	94.5%	<.001	<.001
Frequently seeks verbal reassurance from significant others	Object 2_5	25.7%	12.5%	65.2%	32.9%	21.5%	16.0%	79.4%	<.001	<.001
Reacts negatively when contact with significant others is lacking	Emotion 2_1	34.0%	24.8%	52.6%	24.4%	18.6%	11.8%	77.4%	<.001	<.001
Enjoys activities with significant others	Emotion 2_2	64.2%	46.5%	86.6%	55.7%	46.5%	41.2%	49.5%	<.001	<.001
Dislikes being alone and seeks contact with (significant) others	Emotion 2_3	38.7%	18.8%	65.5%	27.8%	15.1%	23.5%	77.6%	<.001	<.001
Emotional states can be influenced by attention from caregivers	Emotion 2_4	61.9%	49.5%	84.1%	52.8%	39.5%	41.2%	51.6%	<.001	<.001
Is noticeably tense in unfamiliar/confusing social situations	Emotion 2_5	50.7%	52.5%	68.1%	36.9%	34.9%	23.5%	60.0%	.007	<.001
Engages in the same activities as peers, but independently of them (i.e., 'parallel play'—playing side by side rather than with each other)	Peers 2_1	36.3%	25.0%	75.7%	27.0%	9.4%	9.5%	80.3%	<.001	<.001
Is curious about peers and interacts with them briefly	Peers 2_2	40.0%	8.8%	76.2%	45.0%	27.3%	23.8%	75.2%	<.001	<.001
Tries to find out more about peers by watching, listening and/or touching	Peers 2_3	42.0%	24.4%	79.6%	32.4%	23.0%	28.6%	73.8%	<.001	<.001

TABLE 3 (Continued)

SED-S 2 items	Domain and item number	Frequency rate of overall 'yes' answers	Relative frequency rate of 'yes' answers from participants in the respective domain stage					Chi ² -test SED-S 2 versus 1	Chi ² -test SED-S 2 versus 3	
			SED-S 1	SED-S 2 (sensitivity)	SED-S 3	SED-S 4	SED-S 5	Specificity rate	p-value	
Imitates peers when interacting with others										
Engages with peers when authority figures are present	Peers 2_4	14.7%	1.9%	30.9%	15.3%	6.5%	23.8%	92.1%	<.001	.003
Manipulates objects repetitively for sustained periods of time (e.g., putting them into/taking them out of boxes, moving them back and forth, throwing them, etc.)	Peers 2_5	34.0%	6.9%	60.2%	40.5%	25.2%	38.1%	77.0%	<.001	.001
	Material 2_1	21.7%	27.5%	64.1%	26.5%	6.1%	5.6%	84.5%	<.001	<.001
Interacting with significant others is the main focus of activities										
Enjoys interactions that involve repeatedly throwing, dropping and handing objects back and forth	Material 2_2	21.2%	18.7%	57.7%	25.8%	11.6%	7.3%	84.1%	<.001	<.001
	Material 2_3	10.9%	15.4%	35.9%	9.0%	4.3%	3.2%	92.7%	.002	<.001
	Material 2_4	11.6%	14.3%	29.5%	16.1%	3.0%	4.0%	91.0%	.016	.017
Reaches for things he/she can see or hear	Material 2_5	30.9%	50.5%	71.8%	32.3%	13.4%	12.1%	75.1%	.005	<.001
Frequently clings to significant others	Communication 2_1	19.1%	17.4%	40.6%	18.8%	9.3%	2.5%	86.6%	<.001	<.001
Seeks constant physical and verbal contact with significant others by remaining close by their side	Communication 2_2	23.4%	12.8%	53.1%	21.8%	13.5%	5.0%	84.5%	<.001	<.001
Points at things to get others' attention	Communication 2_3	25.5%	19.8%	70.3%	18.2%	9.8%	0.0%	86.4%	<.001	<.001
Repeats sounds, words and short sentences he/she hears										
Uses his/her entire body to get his/her message across	Communication 2_4	20.3%	14.0%	50.0%	19.4%	8.3%	0.0%	87.6%	<.001	<.001
	Communication 2_5	42.8%	62.8%	85.2%	37.0%	18.7%	5.0%	68.4%	<.001	<.001
Negative emotions are regulated when the cause of stress is eliminated	Affect 2_1	57.4%	61.5%	84.6%	50.8%	33.8%	12.9%	51.4%	<.001	<.001
Turns to significant others for protection and consolation	Affect 2_2	46.7%	21.3%	71.1%	49.6%	41.2%	19.4%	61.1%	<.001	<.001
Is able to control (aggressive) impulses if significant others are nearby and ready to intervene if necessary	Affect 2_3	44.8%	30.3%	75.8%	43.0%	22.1%	16.1%	65.2%	<.001	<.001
Is upset and angry when significant others leave	Affect 2_4	12.6%	8.2%	32.9%	6.2%	4.4%	0.0%	94.0%	<.001	<.001
Frustration leads to anger that can be manifested in aggressive behaviour towards significant others, physical restlessness, screaming, hitting or throwing objects	Affect 2_5	42.2%	41.8%	60.4%	43.8%	13.2%	6.5%	63.7%	.002	.001
Mean		31.9%	26.8%	64.7%	29.1%	17.3%	13.2%	77.1%		

Note: Sensitivity and specificity rates >70% are marked in bold. In each SED-S 1 domain, the item with the highest specificity rate and a p-value below .05 is marked in bold.

TABLE 4 Overview of validity assessment for SED-S 3 items.

SED-S 3 items	Domain and item number	Frequency rate of overall 'yes' answers	Relative frequency rate of 'yes' answers from participants in the respective domain stage					Chi ² -test SED-S 3 versus 2	
			SED-S 1	SED-S 2	SED-S 3 (sensitivity)	SED-S 4	SED-S 5	Specificity rate	p-value
			1	2	3	4	5		
Wants to do everything him-/herself (e.g., personal hygiene, meals, etc.)	Body 3_1	47.5%	10.7%	20.5%	63.5%	51.6%	45.2%	64.5%	<.001
Seeks help from others or utilises objects to overcome physical limitations (uses a chair to reach the cookie jar, for example)	Body 3_2	62.3%	20.0%	41.1%	81.4%	61.0%	59.5%	52.1%	<.001
Tries to assert his/her will by means of provocative behaviour (e.g., stripping off clothes, stomping, flinging him-/herself on the ground)	Body 3_3	42.8%	34.7%	50.7%	60.8%	20.8%	14.3%	70.8%	<.001
Uses language (possibly supported by gestures) to communicate	Body 3_4	71.9%	21.3%	28.8%	90.1%	86.8%	66.7%	41.8%	<.001
Goes to the toilet on his/her own, but needs help with hygiene	Body 3_5	42.3%	34.7%	32.9%	62.7%	25.2%	9.5%	73.1%	<.001
Insists on getting his/her way	Others 3_1	68.8%	37.1%	64.5%	91.0%	56.3%	34.8%	45.0%	<.001
Wants attention and tests limits at the same time	Others 3_2	49.8%	10.0%	46.4%	72.6%	37.8%	26.1%	64.3%	<.001
Is open to alternative suggestions	Others 3_3	60.9%	15.7%	53.6%	74.4%	70.6%	65.2%	47.4%	<.001
Only obeys rules when authority figures are present	Others 3_4	31.7%	18.6%	22.3%	51.7%	16.8%	13.0%	80.7%	<.001
Persistently says 'No!' in order to assert his/her will	Others 3_5	37.7%	7.1%	24.7%	62.4%	27.7%	26.1%	77.5%	<.001
Is upset at loss of transitional objects (i.e., emotionally charged 'comfort' objects)	Object 3_1	21.7%	8.3%	14.5%	58.8%	20.2%	12.5%	84.3%	<.001
Transitional/comfort objects provide a sense of security	Object 3_2	24.0%	9.7%	14.5%	61.2%	23.6%	14.6%	82.0%	<.001
Intentionally looks for things and people that can no longer be seen/heard	Object 3_3	66.8%	18.1%	47.8%	88.2%	78.1%	68.8%	36.6%	.042
Emotional bonds are maintained over a distance (e.g., over the telephone)	Object 3_4	58.8%	9.7%	42.0%	74.1%	68.2%	66.7%	43.6%	.306
Shows resentment after being separated from significant others for longer periods of time	Object 3_5	11.9%	1.4%	11.6%	31.8%	9.5%	9.7%	91.3%	<.001
Expresses anger when he/she doesn't get his/her way	Emotion 3_1	59.5%	46.5%	55.6%	84.7%	38.4%	35.3%	50.7%	<.001
Wants caregivers to him-/herself, shows jealousy	Emotion 3_2	28.3%	9.9%	23.7%	50.6%	17.4%	23.5%	80.7%	<.001
Shows defiance towards significant others	Emotion 3_3	55.1%	26.7%	51.7%	83.0%	45.3%	29.4%	56.2%	<.001
Is able to name own basic feelings (e.g., anger, sadness, fear, and happiness)	Emotion 3_4	45.9%	6.9%	29.3%	73.9%	74.4%	70.6%	65.4%	.923
Wants to be the centre of attention	Emotion 3_5	31.9%	4.0%	23.7%	57.4%	32.6%	41.2%	78.4%	<.001
Tries to impose his/her will on peers	Peers 3_1	34.3%	6.9%	27.1%	83.8%	34.5%	42.9%	76.6%	<.001

TABLE 4 (Continued)

SED-S 3 items	Domain and item number	Frequency rate of overall 'yes' answers	Relative frequency rate of 'yes' answers from participants in the respective domain stage					Chi ² -test SED-S 3 versus 2	
			SED-S 1	SED-S 2	SED-S 3 (sensitivity)	SED-S 4	SED-S 5	Specificity rate	p-value
Is reluctant to share materials with peers	Peers 3_2	21.4%	10.0%	24.3%	56.8%	5.0%	4.8%	86.4%	<.001
Shows no regard for what peers want	Peers 3_3	31.5%	10.0%	34.8%	73.9%	22.3%	4.8%	77.8%	<.001
Is jealous of peers	Peers 3_4	22.5%	6.3%	19.9%	49.5%	23.7%	19.0%	83.4%	<.001
Bosses peers around	Peers 3_5	25.8%	2.5%	14.9%	75.7%	25.9%	33.3%	85.2%	<.001
Shares only reluctantly or at the request of significant others	Material 3_1	25.0%	6.6%	23.1%	56.8%	15.9%	12.1%	85.8%	<.001
Engages in activities requiring fine motor skills, such as cutting and pasting	Material 3_2	56.4%	9.9%	26.9%	76.8%	68.9%	66.9%	50.5%	.115
The activity itself is more important than the end result (has no interest in keeping drawings once they're finished, for example)	Material 3_3	40.7%	13.2%	37.2%	83.2%	36.0%	16.1%	73.7	<.001
Frequently examines objects which may then fall apart	Material 3_4	14.2%	9.9%	14.1%	27.7%	9.8%	6.5%	90.4%	<.001
Does activities in which imitation plays a role	Material 3_5	18.3%	2.2%	6.4%	25.8%	22.0%	23.4%	84.2%	.419
Often says 'no' to try to get his/her way	Communication 3_1	38.6%	4.7%	35.9%	67.3%	34.7%	20.0%	72.0%	<.001
Communication is self-centred	Communication 3_2	52.8%	14.0%	43.8%	87.3%	54.4%	15.0%	60.0%	<.001
Communication is mostly related to the immediate situation	Communication 3_3	34.2%	11.6%	29.7%	64.2%	27.5%	5.0%	77.0%	<.001
Uses 'bad' words to provoke reactions from others	Communication 3_4	19.4%	2.3%	7.0%	33.3%	22.3%	25.0%	85.7%	.019
Says things with no regard for the social context	Communication 3_5	36.9%	3.5%	17.2%	61.8%	46.1%	25.0%	72.3%	.003
Responds with aggression when limits are imposed on his/her will	Affect 3_1	49.3%	32.8%	39.6%	73.6%	27.9%	19.4%	66.5%	<.001
Is more willing to cooperate when offered a choice between two alternatives	Affect 3_2	47.4%	18.9%	38.3%	68.2%	52.9%	29.0%	66.2%	.020
Frustration finds expression in physical restlessness, temper tantrums and stubbornly oppositional behaviour	Affect 3_3	59.5%	32.0%	52.3%	86.4%	45.6%	22.6%	58.1%	<.001
Frequently conveys protest using physical or verbal means	Affect 3_4	57.0%	31.1%	42.3%	85.1%	51.5%	22.6%	61.4%	<.001
Is rarely able to talk about the causes and effects of his/her aggressive behaviour	Affect 3_5	36.9%	9.0%	25.5%	65.3%	23.5%	9.7%	81.6%	<.001
Mean		41.0%	14.7%	31.5%	66.9%	37.8%	28.9%	69.5%	

Note: Sensitivity and specificity rates >70% are marked in bold. In each SED-S 1 domain, the item with the highest specificity rate and a p-value below .05 is marked in bold.

TABLE 5 Overview of validity assessment for SED-S 4 items.

SED-S 4 items	Domain and item number	Frequency rate of overall 'yes' answers	Relative frequency rate of 'yes' answers from participants in the respective domain stage						Chi ² -test SED-S 4 versus 3	Chi ² -test SED-S 4 versus 5
			SED-S 1	SED-S 2	SED-S 3	SED-S 4 (sensitivity)	SED-S 5	Specificity rate		
			1	2	3		5	p-value		
Seeks role models of the same gender to identify with (e.g., police officers, firefighters, pop stars, and caregivers)	Body 4_1	27.3%	2.7%	4.1%	17.5%	64.8%	31.0%	85.9%	<.001	<.001
Has developed a gender identity	Body 4_2	62.9%	6.7%	16.4%	66.5%	96.9%	92.9%	49.0%	<.001	.238
Shows a sense of shame/modesty (closes the door when using the toilet, for example)	Body 4_3	39.4%	2.7%	1.4%	29.3%	80.5%	78.6%	75.1%	<.001	.780
Seeks to imitate role models in terms of appearance (e.g., by adopting hair and clothing styles, necklaces, bracelets, etc.)	Body 4_4	17.5%	2.7%	1.4%	8.7%	47.2%	14.3%	92.9%	<.001	<.001
Wants to choose clothing according to personal taste, regardless of the weather or occasion	Body 4_5	21.7%	8.0%	4.1%	20.9%	40.3%	11.9%	84.8%	<.001	<.001
Solicits the opinion and seeks the approval of significant others	Others 4_1	38.7%	1.4%	28.3%	37.2%	73.9%	60.9%	69.8%	<.001	.202
Wants to assume the role of authority figures ('deputy caregiver')	Others 4_2	24.5%	0.0%	11.4%	24.8%	57.1%	21.7%	83.4%	<.001	.002
Makes decisions on his/her own and is aware of the immediate consequences (when crossing the street, for example)	Others 4_3	30.4%	4.3%	16.3%	24.8%	68.1%	73.9%	78.7%	<.001	.579
Identifies with role models and attempts to imitate their characteristic behaviour	Others 4_4	15.8%	1.4%	7.2%	15.8%	37.0%	13.0%	89.2%	<.001	.026
Wants to be perceived and treated in accordance with his/her gender	Others 4_5	36.9%	1.4%	25.3%	32.1%	79.8%	56.5%	73.4%	<.001	.016
Can manage transitions between familiar social contexts (e.g., residential and work settings)	Object 4_1	70.6%	30.6%	29.0%	54.1%	90.9%	86.1%	42.7%	<.001	.143
Is able to part with transitional objects in familiar environments	Object 4_2	22.4%	4.2%	5.8%	29.4%	32.6%	18.1%	84.3%	.582	.002
Authority figures provide reassurance in unfamiliar situations	Object 4_3	67.2%	25.0%	49.3%	51.8%	90.1%	67.4%	47.8%	<.001	<.001
Can engage in familiar activities in unfamiliar environments	Object 4_4	62.9%	18.1%	27.5%	42.4%	81.8%	82.6%	49.5%	<.001	.839
Is insecure in unfamiliar environments when no authority figures are present	Object 4_5	46.4%	16.7%	34.8%	36.5%	71.9%	29.9%	70.3%	<.001	<.001
Is afraid of making mistakes or doing something wrong	Emotion 4_1	22.9%	4.0%	12.5%	25.6%	65.1%	35.3%	84.0%	<.001	.022
Is able to empathise with others (tries to console them, for example)	Emotion 4_2	32.2%	4.0%	19.4%	38.6%	84.9%	41.2%	76.4%	<.001	<.001
Shows feelings of guilt	Emotion 4_3	28.4%	7.9%	17.2%	29.0%	80.2%	35.3%	80.0%	<.001	<.001

TABLE 5 (Continued)

SED-S 4 items	Domain and item number	Frequency rate of overall 'yes' answers	Relative frequency rate of 'yes' answers from participants in the respective domain stage					Chi ² -test SED-S 4 versus 3	Chi ² -test SED-S 4 versus 5	
			SED-S 1	SED-S 2	SED-S 3	SED-S 4 (sensitivity)	SED-S 5	Specificity rate	p-value	
Shows a sense of shame/modesty (facial expression)	Emotion 4_4	25.7%	3.0%	12.1%	25.6%	82.6%	58.8%	83.7%	<.001	.029
Is worried he/she won't be able to handle assigned tasks or responsibilities	Emotion 4_5	18.8%	4.0%	6.9%	18.8%	66.3%	29.4%	89.0%	<.001	.005
Has regular, sustained contact with peers	Peers 4_1	29.6%	0.6%	16.0%	18.9%	82.7%	71.4%	86.0%	<.001	.216
Makes friends with certain peers	Peers 4_2	31.5%	1.3%	19.9%	27.9%	78.4%	71.4%	82.2%	<.001	.475
Imitates caregivers when interacting with peers	Peers 4_3	15.7%	0.0%	6.6%	27.9%	34.5%	23.8%	89.9%	.264	.331
Frequently enjoys activities with peers for sustained periods, but is still primarily focused on what he/she wants	Peers 4_4	28.9%	2.5%	20.4%	24.3%	71.2%	47.6%	83.5%	<.001	.031
Voluntarily shares material with peers	Peers 4_5	39.4%	5.6%	37.6%	28.8%	83.5%	76.2%	73.6%	<.001	.414
Approaches peers by play and shared activities	Material 4_1	36.8%	4.4%	12.8%	17.4%	62.2%	66.1%	72.5%	<.001	.491
Likes games involving imagination (playing 'pretend' and make-believe)	Material 4_2	18.0%	0.0%	6.4%	7.7%	37.8%	25.0%	89.3%	<.001	.021
Engages in creative activities such as painting and drawing	Material 4_3	30.7%	0.0%	1.3%	20.0%	72.0%	30.6%	84.4%	<.001	<.001
Shows imagination and creativity by varying colours, shapes, and so forth, when working with materials	Material 4_4	26.0%	1.1%	1.3%	9.7%	63.4%	30.6%	87.7%	<.001	<.001
Follows simple rules (like waiting his/her turn) when playing games	Material 4_5	54.1%	8.8%	19.2%	37.4%	87.2%	86.3%	58.0%	<.001	.822
Continuously asks 'why' questions	Communication 4_1	19.9%	1.2%	2.3%	15.2%	42.0%	30.0%	90.2%	<.001	.159
Asks questions and contributes information in conversations relating to his/her personal experience and daily life	Communication 4_2	33.0%	0.0%	3.1%	23.0%	69.4%	65.0%	83.8%	<.001	.582
Describes his/her emotions in simple terms	Communication 4_3	43.6%	1.2%	9.4%	32.1%	89.6%	70.0%	77.6%	<.001	.001
Can explain in simple terms why he/she is upset, happy, scared, jealous, proud, and so forth	Communication 4_4	36.1%	1.2%	4.7%	22.4%	79.3%	60.0%	83.8%	<.001	.009
Communication is directed at peers as well as significant others	Communication 4_5	45.1%	0.0%	8.6%	41.8%	86.0%	75.0%	73.7%	<.001	.083
Expresses regret and wants to make amends	Affect 4_1	34.0%	6.6%	28.9%	38.8%	70.6%	48.4%	70.6%	<.001	.033
Talks about what he/she regards as good or acceptable behaviour and gives reasons	Affect 4_2	14.9%	1.6%	4.7%	12.8%	63.2%	25.8%	91.2%	<.001	<.001
	Affect 4_3	32.7%	4.9%	22.8%	40.5%	76.5%	32.3%	72.8%	<.001	<.001

(Continues)

TABLE 5 (Continued)

SED-S 4 items	Domain and item number	Frequency rate of overall 'yes' answers	Relative frequency rate of 'yes' answers from participants in the respective domain stage					Specificity rate	Chi ² -test SED-S 4 versus 3	Chi ² -test SED-S 4 versus 5
			SED-S 1	SED-S 2	SED-S 3	SED-S 4 (sensitivity)	SED-S 5			
Can delay gratification of his/her own wishes for the sake of a reward										
Is able to talk about the causes and effects of his/her aggressive behaviour with support from significant others	Affect 4_4	27.0%	0.0%	13.4%	33.1%	75.0%	45.2%	79.0%	<.001	.004
Is a 'sore loser' and refuses to continue playing after losing a game	Affect 4_5	9.3%	0.0%	4.7%	10.7%	29.4%	12.9%	93.2%	<.001	.075
Mean		33.0%	4.7%	14.3%	28.0%	68.7%	48.3%	77.8%		

Note: Sensitivity and specificity rates > 70% are marked in bold. *p*-values smaller than .05 are marked in bold. In each SED-S 1 domain, the item with the highest specificity rate and a *p*-value below .05 is marked in bold.

ranged between 20.6% and 41.0% across stages. Furthermore, the overall sensitivity rate of the items was on average 67.5% and ranged between 64.7% and 71.9% across stages. The number of items with a sensitivity rate above 70% increased per stage, such that SED-S 1 included 17 items (out of 40; 43%) and SED-S 5 included 25 items (out of 40; 63%) with a sensitivity rate above 70%. Special attention should be given to the items with the lowest sensitivity rates, as these described emotional behaviours that were displayed by only a small number of individuals and may therefore not accurately represent the underlying emotional needs of the individuals within this particular stage. For example, SED-S 1 included one item in the domain 'Communication' and SED-S 3 included one item in the domain 'Materials' with a sensitivity rate of only 24.4% and 25.8%, respectively (e.g., Communication1_4; Material 3_5).

Results showed that the overall specificity rate of the items was on average 79.5% and ranged between 69.5% and 87.4% across stages. Results further indicated that all 40 items (100%) in SED-S 1 showed a specificity rate above 70%. Similarly, most items ($n = 37/40$; 92.5%) in SED-S 5 showed a specificity rate above 70%. These findings suggest that the items in the lowest and highest SED-S stages were most effective in identifying participants who were not assigned to the respective domain stage. A possible explanation is that the items in SED-S 1 and SED-S 5 describe more distinctive behavioural manifestations; this can make it easier for informants to determine that the described behaviours were not displayed by the individual. The number of items with a specificity rate above 70% was lowest in SED-S 3, although more than half of the items ($n = 23/40$; 57.5%) still met this cut-off point.

Most items showed good discriminatory power as they received more 'yes' answers from participants in the respective domain stage than from participants in the adjacent domain stage(s). However, our findings did raise some concerns about the discriminatory power of the items in SED-S 4. That is, 16 out of 40 items (40%) could not discriminate between the response patterns of participants in the respective domain stage and adjacent domain stage SED-S 5. Upon closer examination, it became apparent that all domains, except for the domain 'Emotions', included at least one item with insufficient discriminative power. Notably, in the domain 'Peers', 4 out of 5 items and in the domain 'Communication' 3 out of 5 items had insufficient discriminatory power in the demarcation of the SED-S 4 to SED-S 5. At the same time, it should be noted that most items in SED-S 4 ($n = 38$; 95%) had sufficient power to discriminate between the response patterns of participants in the respective domain stage and adjacent domain stage SED-S 3. Likewise, we found several items that could accurately differentiate between the response patterns of participants in the lower domain stage but not with the response patterns of participants in the higher domain stage or vice versa (e.g., Body 3_3 and Emotion 3_4). This information is important to consider when making final judgements about whether item revisions are needed as the quality of an item may also lie within its ability to accurately differentiate in only one direction. Furthermore, it should be noted that creating a scale with perfect discriminatory power is challenging as ED is a dynamic, gradual and continuous process

TABLE 6 Overview of validity assessment for SED-S 5 items.

SED-S 5 items	Domain and item number	Frequency rate of overall 'yes' answers	Relative frequency rate of 'yes' answers from participants in the respective domain stage					Specificity rate	Chi ² -test SED-S 5 versus 4 p-value
			SED-S 1	SED-S 2	SED-S 3	SED-S 4	SED-S 5 (sensitivity)		
Seeks to assert his/her position in the peer group by means of his/her appearance	Body 5_1	9.3%	0.0%	0.0%	3.4%	17.6%	47.6%	93.5%	<.001
Is eager to demonstrate physical prowess and compete with others	Body 5_2	13.4%	0.0%	6.8%	7.6%	21.4%	54.8%	89.6%	<.001
Shows modesty/a sense of shame in respect to sexuality	Body 5_3	24.7%	0.0%	1.4%	18.3%	41.5%	85.7%	79.8%	<.001
Is able to assess his/her physical abilities accurately	Body 5_4	30.2%	8.0%	11.0%	24.7%	42.1%	92.9%	74.4%	<.001
Is concerned with his/her appearance and increasingly able to gauge how he/she is perceived by others realistically	Body 5_5	13.9%	0.0%	1.4%	6.1%	23.3%	73.8%	90.5%	<.001
Is eager to compete with significant others	Others 5_1	10.8%	0.0%	3.6%	12.4%	18.5%	39.1%	90.3%	.028
Conforms to social norms and rules even when no authority figures are present	Others 5_2	35.3%	2.9%	30.1%	23.9%	73.1%	91.3%	66.9%	.610
Models behaviour on peers more than on authority figures	Others 5_3	8.8%	1.4%	2.4%	5.6%	19.3%	56.5%	93.0%	<.001
Looks to individuals outside his/her immediate everyday environment (e.g., athletic coaches, etc.) for guidance	Others 5_4	10.1%	0.0%	5.4%	7.3%	17.6%	65.2%	92.0%	<.001
Seeks to gain approval by taking on responsibilities or tasks and pointing accomplishments out to significant others ('look what I did')	Others 5_5	23.2%	0.0%	13.3%	18.4%	50.4%	73.9%	78.8%	.038
Initiates social activities on his/her own	Object 5_1	32.0%	0.0%	17.4%	11.8%	27.3%	75.0%	81.2%	<.001
Explores unfamiliar environments of his/her own accord	Object 5_2	30.1%	5.6%	10.1%	14.1%	17.8%	81.9%	85.9%	<.001
Adapts to changing situational demands	Object 5_3	49.3%	11.1%	10.1%	12.9%	58.7%	93.1%	64.1%	<.001
Pursues interests beyond his/her familiar environment	Object 5_4	43.5%	5.6%	10.1%	23.5%	43.4%	90.3%	70.9%	<.001
Applies familiar behaviour in new contexts	Object 5_5	50.5%	11.1%	7.2%	21.2%	60.3%	91.7%	62.2%	<.001
Is worried about not being accepted by peers	Emotion 5_1	13.2%	1.0%	4.3%	12.5%	40.7%	76.5%	88.6%	.007
Is concerned about his/her appearance	Emotion 5_2	18.3%	2.0%	8.2%	24.4%	38.4%	88.2%	83.7%	<.001
Strives to win the approval of peers	Emotion 5_3	17.0%	0.0%	8.2%	18.8%	43.0%	88.2%	85.0%	<.001
Is concerned about violating social conventions and rules	Emotion 5_4	10.1%	0.0%	4.3%	8.0%	33.7%	52.9%	91.1%	.133
Reflects on and regulates his/her emotions	Emotion 5_5	9.5%	0.0%	2.6%	8.0%	32.6%	58.8%	91.9%	.040
Maintains close friendships with peers	Peers 5_1	12.4%	0.0%	4.4%	7.2%	30.9%	81.0%	90.0%	<.001
Is competitive with peers	Peers 5_2	10.6%	0.6%	5.5%	10.8%	23.0%	47.6%	90.7%	<.001
Wants to 'belong' and be popular with peers	Peers 5_3	22.4%	0.6%	13.3%	20.7%	50.4%	90.5%	80.0%	<.001
Shows loyalty to friends	Peers 5_4	13.4%	0.6%	3.9%	9.9%	33.1%	81.0%	89.0%	<.001
Consults with peers to find solutions to conflicts/problems	Peers 5_5	7.0%	0.6%	2.8%	2.7%	16.5%	52.4%	94.6%	<.001
Can share materials, tasks and responsibilities in activities with peers	Material 5_1	41.3%	11.0%	10.3%	20.6%	55.5%	90.3%	71.1%	<.001
	Material 5_2	29.7%	0.0%	0.0%	7.1%	40.2%	84.7%	84.2%	<.001

(Continues)

TABLE 6 (Continued)

SED-S 5 items	Domain and item number	Frequency rate of overall 'yes' answers	Relative frequency rate of 'yes' answers from participants in the respective domain stage					Specificity rate	Chi ² -test SED-S 5 versus 4 p-value	
			SED-S 1	SED-S 2	SED-S 3	SED-S 4	SED-S 5 (sensitivity)			
The primary aim of activities is achieving a good result (saves only 'nice' drawings, for example)										
Receiving recognition for the final product/result is important	Material 5_3	35.3%	1.1%	5.1%	15.5%	53.7%	79.8%	76.0%	<.001	
Works with others towards a common goal, for example, in team sports and group activities	Material 5_4	13.4%	0.0%	0.0%	1.9%	14.0%	45.2%	94.7%	<.001	
Participates in group activities of a competitive nature, such as races, bowling, and so forth	Material 5_5	22.2%	0.0%	10.3%	9.0%	17.1%	69.4%	89.8%	<.001	
Asks questions about cause and effect and how things work	Communication 5_1	12.9%	0.0%	0.0%	4.2%	22.3%	72.5%	91.3%	<.001	
Communicates with peers and enjoys exchanges with them	Communication 5_2	28.8%	1.2%	4.7%	18.8%	53.9%	85.0%	75.2%	<.001	
Uses objective/factual arguments to support his/her opinion in discussions	Communication 5_3	15.7%	0.0%	1.6%	8.5%	24.9%	80.0%	88.8%	<.001	
Talks about his/her feelings	Communication 5_4	16.2%	0.0%	0.8%	3.0%	31.1%	82.5%	88.5%	<.001	
Talks about his/her strengths and weaknesses	Communication 5_5	13.2%	0.0%	0.8%	4.2%	21.2%	80.0%	91.4%	<.001	
Aggression is primarily expressed using verbal means	Affect 5_1	28.1%	4.1%	12.1%	36.4%	51.5%	83.9%	74.9%	.002	
Is able to regulate his/her emotions by talking about his/her feelings and needs and seeking support from others in the group	Affect 5_2	4.9%	0.0%	0.7%	2.1%	16.2%	41.9%	97.1%	.006	
Reflects on what did or did not go well without prompting from others	Affect 5_3	13.9%	0.0%	4.7%	10.3%	47.1%	67.7%	89.0%	.056	
Responds with aggression when he/she feels threatened, slighted or extremely frustrated	Affect 5_4	24.0%	4.9%	12.8%	29.3%	45.6%	64.5%	78.1%	.081	
Displays anger or aggression in competitive situations	Affect 5_5	3.6%	0.0%	0.7%	3.7%	8.8%	19.4%	97.2%	.136	
Mean		20.6%	1.8%	6.3%	12.7%	34.4%	71.9%	84.6%		

Note: Sensitivity and specificity rates >70% are marked in bold. *p*-values smaller than .05 are marked in bold. In each SED-S 1 domain, the item with the highest specificity rate and a *p*-value below .05 is marked in bold.

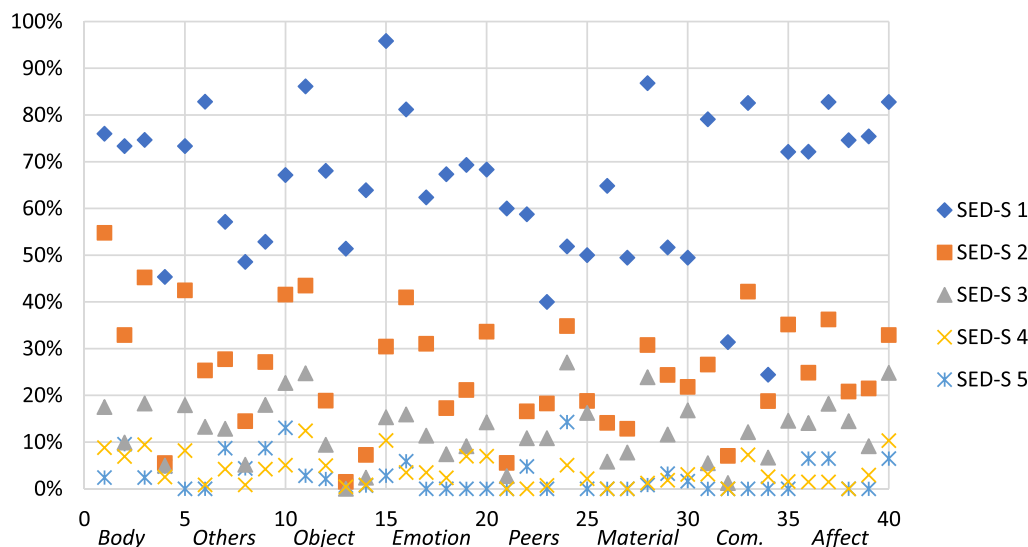


FIGURE 2 SED-S 1: Relative frequency rates of 'yes' answers from participants in the respective domain stage and from participants in a different domain stage. The x-axis shows the 40 items of SED-S 1. The y-axis shows the relative frequency of 'yes' answers from participants in the respective SED-S domain stage. ♦ Corresponds to the item sensitivity rate. Com., Communication.

(Tarasova et al., 2022). It may therefore be expected that individuals display a wide range of emotional behaviours which do not always fit perfectly in one of the five SED-S domain stages. Hence, some overlap in the response patterns of participants in the adjacent domain stages can be expected and requires acceptance.

Prior research demonstrated that SED-S domain scores form a coherent and unidimensional concept (Flachsmeyer et al., 2023; Meinecke & Sappok, 2021; Sterkenburg et al., 2021). Surprisingly, this study found a discrepancy between emotional functioning depicted within seven out of the eight domains and one's overall level of emotional functioning. As such, it seems reasonable to suggest that there may be a need to make some adjustments to the current approach for determining one's overall level of ED. Note that to determine overall level of ED, domain scores are currently ranked from lowest to highest and the fourth lowest domain stage signifies overall level of ED (Sappok et al., 2016). One possible adjustment may be to incorporate the item's specificity rate and discriminatory power in case two domain stages have the same amount of 'yes' answers. For example, a 'yes' answer to an item with the highest specificity rate and discriminatory power within a specific domain stage could be scored double. This may be a more fruitful approach than choosing the lower-level stage when there is a tie between two domain stages, although this new approach requires further investigation. Apart from this, our findings point to the importance of evaluating the domain scores in addition to the overall SED-S score when selecting appropriate and individualised treatment programs.

This was an exploratory study in which we performed a relatively large number of statistical analyses. We did not account for multiple testing as the primary goal of this exploratory study is to identify potential signals rather than to proof specific hypotheses (also suggested by Bender & Lange, 2001; see also Perneger, 1998). It follows

logically that the present study can only draw preliminary conclusions about the psychometric properties of the items on the SED-S. Future studies are needed to confirm our findings and may provide further insight into which items may require removal or revision in order to improve the psychometric properties of the scale.

A major strength of the present study is that it was the first to perform a rigorous evaluation of the item-level properties of the SED-S in a large, multicentre sample of adults with intellectual disabilities. Furthermore, the study was pre-registered on the Open Science Framework (OSF) which maximised the credibility, transparency, reproducibility of our findings and minimised the influence of researcher degrees of freedom. Nonetheless, some limitations should be considered when interpreting the results. First, we included a clinical sample and a relatively large number of participants were diagnosed with both an intellectual disability and a mental disorder. Prior work demonstrated lower levels of ED in participants with intellectual disabilities and ASD (Flachsmeyer et al., 2023; Sterkenburg et al., 2021). Furthermore, a pilot study showed associations between other types of mental disorders (e.g., anxiety disorder, schizophrenia) and level of ED (Lytochkin et al., 2021). Future studies may examine whether item response patterns of people with intellectual disabilities are impacted by co-occurring mental disorders. Second, we included a combined sample consisting of participants from Belgium, Germany, and the Netherlands. Hence, we cannot rule out that cross-cultural differences may have impacted the results. Nevertheless, a prior study reported no significant differences in the distribution of ED levels among participants from these three European countries, highlighting the applicability of the SED-S across different countries and languages (Flachsmeyer et al., 2023). Third, the distribution of participants across the different domain levels was not balanced as a relatively small number of participants were assigned to the domain stages of SED-S

5. Future research may strive to include more participants, with and without an intellectual disability, within the highest domain stage. Finally, the present study focused on the five-stage model and insight into the psychometric properties of the items in the recently added stage (SED-S 6; Tarasova et al., 2022) is awaiting examination.

5 | CONCLUSION

The use of valid and reliable instruments to assess ED is a prerequisite for a proper diagnostic assignment. The appraisal of the level of emotional functioning gives insights into the underlying emotional needs of individuals with intellectual disabilities and supports effective and personalised treatments. This study adds to previous validation studies by showing that most items on the SED-S have psychometrically sound properties. Furthermore, we provided a detailed description of the sensitivity and specificity rates as well as the discriminatory power of all items on the scale. These results should be collectively interpreted when making judgements about whether removal or revision of items is necessary and warranted. Future assenting studies are needed to replicate our findings which can bring together qualitative (e.g., Sappok et al., 2016) and quantitative (e.g., Flachsmeyer et al., 2023; Meinecke & Sappok, 2021; Sterkenburg et al., 2021) research findings to further improve the validity of the assessment of the level of ED.

AUTHOR CONTRIBUTIONS

Hauke Hermann: Data preparation; conception of study design; data analysis; manuscript writing. **Annemieke Witte:** Data analysis; manuscript writing. **Gloria Kempelmann:** Conception of study design; data analysis; revision of manuscript. **Brian F. Barrett:** Data acquisition; revision of manuscript. **Sandra Zaal:** Data acquisition; revision of manuscript. **Jolanda Vonk:** Data acquisition; revision of manuscript. **Filip Morisse:** Data acquisition; revision of manuscript. **Anna Pöhlmann:** Data analysis; revision of manuscript. **Paula S. Sterkenburg:** Data acquisition; conception of study design; revision of manuscript. **Tanja Sappok:** Data acquisition; conception of study design; revision of manuscript.

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CONFLICT OF INTEREST STATEMENT

Tanja Sappok and Brian Barrett received royalties from Hogrefe and Kohlhammer for the publication of the SED-S manual and other book

projects. Jolanda Vonk received royalties from Acco, Pavilion and Boom publishers for the publication of book projects about emotional development. The other authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data is available upon reasonable request.

INFORMED CONSENT

Prior to the start of the study, participants and/or their legal guardians signed for informed consent.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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